# Gaia DR3 4277855016732107520: A High-Priority Spectroscopic Follow-Up Target from the Gaia DR3 Astrometric Orbital Catalogue

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## ABSTRACT

We identify Gaia DR3 4277855016732107520 as a high-priority spectroscopic follow-up target, selected from the Gaia DR3 non-single-star (NSS) catalogue on the basis of its formally high companion mass, bright apparent magnitude, and favourable observational properties. The source carries a purely astrometric orbital solution (model type `Orbital`), which derives the companion mass entirely from the along-scan photo-centre wobble with no independent radial-velocity orbit. All inferred companion properties are therefore *conditional on the Gaia pipeline modelling assumptions*; follow-up studies have shown that $\gtrsim$ 30–50 per cent of such `Orbital` solutions fail independent RV validation.

The visible component has Gaia GSP-Phot parameters $T_{\rm eff} = 5922\,{\rm K}$ and $\log g = 2.93$; at RUWE = 9.31 these values may be systematically biased, and no independent spectroscopic classification exists. We adopt a provisional $M_1 = 1.34 \pm 0.40\,{\rm M}_\odot$ from isochrone matching of the GSP-Phot parameters. Conditional on the Gaia orbital solution, a Monte Carlo posterior yields a median companion mass $M_2 = 12.3\,{\rm M}_\odot$ (68 per cent CI [7.1, 22.7] $\rm M_\odot$; 90 per cent CI [5.0, 35.9] $\rm M_\odot$) after applying a $\times 1.7$ parallax-error inflation. This posterior is a heuristic working estimate: the inflation factor is calibrated at the population level, correlated parameters are sampled independently (the full NSS covariance matrix is unavailable), and the SED model is approximate.

Using atmosphere-corrected synthetic flux ratios, a main-sequence star at the inferred mass would exceed the primary flux by $\sim$29–55$\times$ in optical bands, inconsistent with the observed single-star SED. Of seven non-BH scenarios evaluated conditional on the adopted solution, three are photometrically inconsistent (MS companion, WD, NS), one is strongly disfavoured (triple), one is not favoured (stripped He star), and two — astrometric artefact and chance alignment — *cannot presently be excluded*.

The source merits priority follow-up because: (i) the predicted RV semi-amplitude ($K_1 \approx 65\,{\rm km\,s^{-1}}$) is large enough for efficient ground-based confirmation with 2–4 m class facilities; (ii) at $G = 11.25$ and Dec = $+3.5°$, it is accessible from both hemispheres; and (iii) the conditional mass estimate places it in a regime where, if the orbital solution is genuine, the companion would exceed the canonical BH mass threshold ($P(M_2 > 5\,{\rm M}_\odot) \approx 95$ per cent under the adopted assumptions). Until RV confirmation is obtained, the companion mass and nature remain entirely provisional.

**Key words:** binaries: visual – astrometry – stars: fundamental parameters – stars: individual: Gaia DR3 4277855016732107520

## 1 INTRODUCTION

Theoretical models predict $10^8$–$10^9$ stellar-mass black holes (BHs) in the Milky Way (Agol & Kamionkowski 2002; Shapiro & Teukolsky 1983), yet fewer than $\sim$70 are dynamically confirmed, nearly all in X-ray binaries undergoing active accretion (Remillard & McClintock 2006). The vast majority are expected to reside in non-interacting (dormant) systems, producing no electromagnetic signature beyond gravitational influence on a luminous companion.

Gaia Data Release 3 (DR3; Gaia Collaboration et al. 2023a) has transformed the search for dormant BHs. The non-single-star (NSS) catalogue (Gaia Collaboration et al. 2023b) provides orbital solutions for $\sim$180 000 binaries, including astrometric orbits that — under the modelling assumptions of the pipeline — yield direct companion mass estimates rather than the minimum masses available from spectroscopic (SB1) solutions alone. These solutions come in several model types, ranging from the purely astrometric

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`Orbital` class (photo-centre orbit only) to `AstroSpectroSB1` solutions that combine astrometry with an independent spectroscopic orbit. The latter provide substantially stronger constraints because the radial-velocity orbit independently confirms the period, eccentricity, and semi-amplitude. By contrast, an `Orbital` solution derives the companion mass entirely from the along-scan astrometric residuals and is therefore more vulnerable to pipeline modelling errors and scan-geometry biases.

The landmark discoveries of Gaia BH1 ($M_2 \approx 9.6\,\mathrm{M}_\odot$; El-Badry et al. 2023a), Gaia BH2 ($M_2 \approx 8.9\,\mathrm{M}_\odot$; El-Badry et al. 2023b), and Gaia BH3 ($M_2 \approx 33\,\mathrm{M}_\odot$; Gaia Collaboration et al. 2024) have established the efficacy of astrometric BH detection. Crucially, each of these three confirmed systems benefited from independent multi-epoch radial-velocity follow-up that validated the astrometric orbit — a step that has not yet been performed for the present source. Systematic searches have since expanded the candidate list (Jayasinghe et al. 2021; Thompson et al. 2019; Shenar et al. 2022; Mahy et al. 2022), but El-Badry et al. (2024) have demonstrated that a substantial fraction of high-significance `Orbital` solutions in the NSS catalogue fail independent RV checks, underscoring the need for spectroscopic confirmation of each individual candidate.

In this paper we present a single-object archival characterization and follow-up prioritization of Gaia DR3 4277855016732107520, selected from the Gaia DR3 NSS catalogue on the basis of its formally high companion mass ($M_2 \simeq 12.3\,\mathrm{M}_\odot$ under pipeline assumptions), bright apparent magnitude ($G = 11.25$), and favourable observing geometry. The system is a wide binary ($P = 424\,\mathrm{d}$) with a dark companion whose mass from the astrometric orbit is $M_2 \simeq 12.3\,\mathrm{M}_\odot$ under Gaia NSS modelling assumptions. Located at $\sim 657\,\mathrm{pc}$ in Ophiuchus ($l = 32.1^\circ$, $b = 9.3^\circ$), the system has moderate extinction, a parallax with $\sim 10$ per cent relative uncertainty, and is accessible from both hemispheres.

This source is a particularly attractive follow-up target for three reasons. First, the predicted RV semi-amplitude ($K_1 \approx 65\,\mathrm{km\,s^{-1}}$) ranks 7th out of 47 `Orbital` BH candidates and 1st among those brighter than $G = 12$ (Section 6.3), making confirmation efficient with 2–4 m class echelle spectrographs. Second, the bright apparent magnitude ($G = 11.25$) and moderate extinction keep the required exposure times short. Third, the equatorial declination (Dec $= +3.5^\circ$) makes the source accessible from both hemispheres year-round. These observational advantages motivate presenting this source as a concrete observing-proposal target rather than waiting for Gaia DR4.

The specific contribution of this paper is a single-object follow-up dossier: (i) the first comprehensive multi-wavelength compilation for this source (Gaia, 2MASS, WISE, GALEX, ROSAT, VLASS, RACS); (ii) a Monte Carlo companion-mass posterior that propagates all identified uncertainties, explicitly flagged as a conditional working estimate; (iii) a systematic evaluation of seven non-BH scenarios with conditional verdicts; and (iv) quantitative follow-up predictions (RV semi-amplitude, epoch count, telescope class) usable for proposal planning. We also provide a quantitative comparison against the broader `Orbital` BH-candidate population (Section 6.3) to justify highlighting this particular source.

We stress from the outset that the solution type is `Orbital` (astrometry only): *no independent spectroscopic orbit exists*. The companion mass therefore depends entirely on the Gaia pipeline model, the parallax, and the adopted primary-mass estimate. Until independent multi-epoch radial velocities confirm or refute the orbital parameters, this system must be treated as a *candidate* rather than a confirmed compact object, and the exclusion of alternative scenarios is itself conditional on the validity of the underlying `Orbital` solution.

The paper is structured as follows. Section 2 describes the observational data. Section 3 presents the analysis methodology. Section 4 reports results. Section 5 evaluates non-BH explanations, conditioned on the adopted Gaia solution. Section 6 discusses implications, the evidence hierarchy, the reliability of the Gaia `Orbital` solution, and follow-up strategy, and Section 7 summarises our findings.

## 2 DATA

### 2.1 Gaia DR3 astrometry and NSS orbital solution

Gaia DR3 4277855016732107520 appears in both the main astrometric catalogue and the NSS orbital supplement with a purely astrometric orbital solution (model type: `Orbital`). This is a critical distinction: unlike `AstroSpectroSB1` solutions (which combine astrometry with a spectroscopic orbit), the `Orbital` model derives the companion mass solely from the photo-centre wobble, making it more susceptible to scan-geometry artefacts and modelling systematics.

Key astrometric parameters include parallax $\varpi = 1.5228 \pm 0.1549\,\mathrm{mas}$ (distance $d \approx 657\,\mathrm{pc}$; 10.2 per cent relative error), and photometry $G = 11.25$, $G_{\mathrm{BP}} = 11.66$, $G_{\mathrm{RP}} = 10.66$ ($G_{\mathrm{BP}} - G_{\mathrm{RP}} = 0.993$). The renormalised unit-weight error RUWE$= 9.31$ is extremely elevated, as expected for an unresolved binary with large photo-centre motion — but also a regime where Gaia GSP-Phot stellar parameters become unreliable (Andrae et al. 2023). The astrometric excess noise significance is $3753\sigma$.

The NSS orbital solution yields: period $P = 424.403 \pm 1.159\,\mathrm{d}$, eccentricity $e = 0.3427 \pm 0.0194$, and solution significance $75.4\sigma$. Because no full NSS covariance matrix is published, we are unable to assess parameter correlations directly; our Monte Carlo approach (Section 3.6) independently samples each parameter, which may underestimate the true uncertainty if correlations are significant.

The Gaia DR3 GSP-Phot pipeline provides stellar parameters: $T_{\mathrm{eff}} = 5922\,\mathrm{K}$ and $\log g = 2.93$. We caution that at RUWE $= 9.31$, GSP-Phot parameters may be systematically biased by the unresolved orbital motion; the photometric evolved-star classification should therefore be regarded as a *working assumption* pending independent spectroscopic confirmation (Section 2.4). The systemic radial velocity is $v_{\mathrm{sys}} = -29.49 \pm 0.57\,\mathrm{km\,s^{-1}}$.

### 2.2 Multi-wavelength photometry

We compile multi-band photometry from 2MASS ($J = 9.84$, $H = 9.42$, $K_s = 9.33$; Cutri et al. 2003) and AllWISE ($W1 = 9.23$, $W2 = 9.27$; Cutri et al. 2014). WISE colours ($W1 - W2 = -0.04$) show no mid-IR excess, consistent with a stellar photosphere. GALEX (Bianchi et al. 2017) provides a near-UV detection (NUV $= 17.52$) but no far-UV counterpart,

consistent with chromospheric emission from an evolved star without a hot companion.

### 2.3 X-ray and radio archival data

We query the ROSAT All-Sky Survey (2RXS; Boller et al. 2016) and XMM-Newton serendipitous catalogue (4XMM-DR13; Webb et al. 2020) within 30″ and 15″ respectively, finding no counterpart. The ROSAT non-detection implies an order-of-magnitude archival upper limit of $L_X \lesssim 5 \times 10^{30}\,\mathrm{erg\,s^{-1}}$, based on the typical RASS sensitivity at the source position and distance, assuming a representative soft X-ray spectrum and negligible additional absorption correction. This limit is consistent with a non-accreting system.

At radio wavelengths, we find no counterpart in the VLA Sky Survey (VLASS; Lacy et al. 2020) within 5″ ($3\sigma < 360\,\mu$Jy at 3 GHz; the source at Dec = +3.5° is within the VLASS footprint) or in the Rapid ASKAP Continuum Survey (RACS; McConnell et al. 2020) within 10″ ($3\sigma < 750\,\mu$Jy at 887.5 MHz).

The source lies at Galactic longitude $l = 32.1°$, placing it in the eastern Galactic hemisphere assigned to the eROSITA-RU consortium. The publicly available eROSITA-DE eRASS1 catalogue (Merloni et al. 2024) covers only the western hemisphere ($l = 180$–$360°$), so no eROSITA data are available for this position.

### 2.4 SIMBAD classification

SIMBAD (Wenger et al. 2000) returns only a Tycho-2 match (TYC 456-894-1) classified as "Star" with no spectral type, variability flag, or prior binarity designation. No independent spectroscopic classification exists. The primary's evolutionary state therefore rests entirely on Gaia GSP-Phot photometric parameters, which are known to be unreliable at high RUWE (Section 2.1).

## 3 METHODS

### 3.1 Gaia NSS mass derivation chain

The companion mass $M_2$ reported for this source originates from two distinct Gaia DR3 tables. The `gaiadr3.nss_two_body_orbit` table provides the astrometric orbital elements (period, eccentricity, angular semi-major axis, inclination, etc. ) fitted to the along-scan residuals. The `gaiadr3.binary_masses` table then combines these elements with the parallax and an external primary-mass estimate to derive $M_2$ via Kepler's third law. The primary mass entering the `binary_masses` derivation is itself model-dependent: it relies on Gaia GSP-Phot atmospheric parameters and isochrone-based mass estimation.

Hence, while the `Orbital` solution eliminates the $\sin i$ degeneracy inherent to SB1 orbits — a critical advantage — the resulting $M_2$ is not a purely empirical "true mass" but depends on the modelling assumptions of the NSS pipeline, the parallax, and the adopted primary mass. We therefore quote the catalogue value $M_2 = 12.31\,\mathrm{M_\odot}$ (with formal uncertainty $\sigma_{M_2} = 7.52\,\mathrm{M_\odot}$) as a *derived mass under Gaia modelling assumptions* and propagate the relevant uncertainties independently via Monte Carlo (Section 3.6).

The large formal catalogue uncertainty ($\sigma_{M_2}/M_2 \approx 61$ per cent) itself reflects the dominance of the parallax uncertainty when propagated correctly through Kepler's third law, as we demonstrate in Section 4.4.

El-Badry et al. (2024) have shown that a substantial fraction of extreme-mass-ratio candidates in the Gaia NSS catalogue are spurious when tested against independent radial velocities, particularly among `Orbital`-type solutions. The present source has *not* been flagged by any published vetting study, but the same failure modes could in principle apply. Independent spectroscopic confirmation therefore remains essential (Section 6.6).

### 3.2 Orbital dynamics

The total system mass follows from Kepler's third law:

$$M_{\rm tot} = M_1 + M_2 = \frac{4\pi^2\, a^3}{G\, P^2}\,, \tag{1}$$

where the true semi-major axis $a$ derives from the angular semi-major axis and parallax. Because $a \propto \varpi^{-1}$, the total mass scales as $M_{\rm tot} \propto \varpi^{-3}\, P^{-2}$, making it extremely sensitive to parallax uncertainty: a 10 per cent parallax error maps to a $\sim$33 per cent mass error. This steep dependence is the dominant source of uncertainty in our posterior (Section 4.4).

### 3.3 Primary-mass estimation

The primary mass is estimated from the Gaia GSP-Phot parameters ($T_{\rm eff} = 5922\,\mathrm{K}$, $\log g = 2.93$). We note that GSP-Phot effective temperatures for RGB stars carry known systematic uncertainties of $\sim$100–200 K (Andrae et al. 2023), and that at RUWE = 9.31 the parameters may be additionally biased by unresolved orbital motion. No independent spectroscopic classification exists (Section 2.4), so the evolved-star identification rests entirely on pipeline photometric parameters. We adopt $M_1 = 1.34 \pm 0.40\,\mathrm{M_\odot}$, assigning a 30 per cent fractional uncertainty to encompass age, metallicity, and systematic errors; the sensitivity of our results to this choice is explored in Section 4.5.

### 3.4 Extinction and SED self-consistency

The source lies at moderate Galactic latitude ($b = 9.3°$), where extinction is present but manageable. We derive the colour excess from the discrepancy between the observed $G_{\rm BP} - G_{\rm RP} = 0.993$ and the intrinsic colour for $T_{\rm eff} = 5922\,\mathrm{K}$ ($(G_{\rm BP} - G_{\rm RP})_0 \approx 0.76$):

$$E(G_{\rm BP} - G_{\rm RP}) = 0.993 - 0.76 = 0.233\,\mathrm{mag}\,, \tag{2}$$

yielding $E(B-V) = 0.175$ mag and $A_V = 0.543$ mag via the Cardelli et al. (1989) extinction law with $R_V = 3.1$.

We note a self-consistency tension in the primary's physical parameters. The *photometric* estimate (from the distance modulus, extinction-corrected bolometric magnitude, and a bolometric correction $BC_G \approx -0.3$ mag) yields $R_\star = 4.5\,\mathrm{R_\odot}$ and $L_\star = 22\,\mathrm{L_\odot}$. However, combining the adopted primary mass with the GSP-Phot surface gravity via $R = \sqrt{G\,M_1/g}$ gives a *logg-based* estimate of $R_\star = 6.5\,\mathrm{R_\odot}$ and $L_\star = 47\,\mathrm{L_\odot}$ — a factor $\sim$1.5× larger in radius and $\sim$2.1× in luminosity. This discrepancy may reflect the unreliability of $\log g$ at RUWE = 9.31 (Andrae et al. 2023), where unresolved orbital

motion degrades the photometric solution. We adopt the photometric values ($R = 4.5\,\mathrm{R}_\odot$, $L = 22\,\mathrm{L}_\odot$) as our fiducial estimates but report both throughout and flag the discrepancy as a systematic uncertainty to be resolved by independent spectroscopy.

### 3.5 SED fitting

We construct the spectral energy distribution from 8 photometric bands (Gaia $G$, $G_{\mathrm{BP}}$, $G_{\mathrm{RP}}$; 2MASS $J$, $H$, $K_s$; WISE $W1$, $W2$) after applying band-dependent extinction corrections. The dereddened SED is compared to a single-temperature model at $T_{\mathrm{eff}} = 5922\,\mathrm{K}$. Residuals below 5 per cent in all bands show that the photometry is consistent with a single luminous star and no detectable secondary light contribution.

For the companion-exclusion analysis, we cross-check the Planck-based flux ratios against atmosphere-corrected values. Wavelength-dependent correction factors are derived from published temperature–bolometric-correction grids (ATLAS9/MARCS and PHOENIX; Castelli & Kurucz 2003; Husser et al. 2013), applied as multiplicative offsets to the Planck ratio in each photometric band. The atmosphere-corrected factors differ from the pure Planck ratios by 2–13 per cent (Table 2), confirming that the MS companion exclusion is robust at the order-of-magnitude level independent of the detailed atmospheric model adopted. For the stripped helium star scenario, the atmosphere correction increases the predicted NUV contrast from $\sim 1360\times$ to $\sim 1660\times$, reinforcing the NUV-based exclusion.

### 3.6 Monte Carlo mass posterior

We draw $N = 200\,000$ Monte Carlo realisations of the companion mass, propagating uncertainties in:

(i) **Parallax** $\varpi$: Gaussian ($1.5228 \pm 0.1549\,\mathrm{mas}$), with an additional $\times 1.7$ inflation factor applied to $\sigma_\varpi$ to account for the known underestimation of external parallax errors for `Orbital`-type solutions at $G \approx 11$ (El-Badry et al. 2024).

(ii) **Period** $P$: Gaussian ($424.403 \pm 1.159\,\mathrm{d}$).

(iii) **Primary mass** $M_1$: log-normal prior centred on $1.34\,\mathrm{M}_\odot$ with $\sigma = 0.40\,\mathrm{M}_\odot$.

(iv) **Model systematic**: uniform multiplicative factor [0.95, 1.05] on $M_{\mathrm{tot}}$.

The total mass scales as (Section 3.2):

$$M_{\mathrm{tot}} \propto \varpi^{-3}\, P^{-2}\,, \tag{3}$$

so the parallax uncertainty dominates. We emphasise that the $\times 1.7$ inflation is an *approximate* correction; the true external-to-internal error ratio for this specific source is unknown. We adopt $\times 1.7$ as the fiducial external-to-internal parallax inflation following published validation work for `Orbital` solutions in this magnitude regime (El-Badry et al. 2024), while Table 4 shows that our conclusions remain sensitive to the exact inflation factor assumed. We report posteriors both with and without inflation (Section 4.4).

Because $M_2$ is derived from the full astrometric orbit (not a minimum mass), no inclination marginalisation is needed. However, the resulting mass depends on all assumptions entering the Gaia NSS pipeline (Section 3.1). The Monte Carlo draws do not account for potential systematic errors in the orbital model itself (e.g. incorrect inclination from scan-geometry biases) — only for the quoted parameter uncertainties.

We emphasise that this posterior represents a *conditional working estimate* of the companion mass under the adopted assumptions, not a rigorously grounded Bayesian inference. In particular: (i) the $\times 1.7$ inflation is calibrated at the population level, not for this individual source; (ii) the independent sampling of correlated parameters (necessitated by the unavailability of the full NSS covariance matrix) may underestimate the true uncertainty; and (iii) the uniform $[0.95, 1.05]$ multiplicative systematic is an ad hoc bracket whose range is not derived from first principles. A fully rigorous uncertainty model would require access to the epoch-level astrometric data and the full parameter covariance matrix, neither of which is publicly available in DR3.

### 3.7 Companion-light exclusion

For a hypothetical main-sequence companion of mass $M_c$, we compute the expected luminosity using empirical mass–luminosity relations ($L \propto M^{3.5}$, $R \propto M^{0.57}$) and compare the resulting *band-by-band* flux ratio using Planck functions:

$$\frac{F_{\lambda,\,\mathrm{comp}}}{F_{\lambda,\,\mathrm{prim}}} = \left(\frac{R_{\mathrm{comp}}}{R_{\mathrm{prim}}}\right)^2 \frac{B_\lambda(T_{\mathrm{comp}})}{B_\lambda(T_{\mathrm{prim}})}\,. \tag{4}$$

As a practical heuristic, we consider a companion detectable if its flux exceeds 5 per cent of the primary in *any* photometric band. We perform this test using both the photometric ($R = 4.5\,\mathrm{R}_\odot$) and logg-based ($R = 6.5\,\mathrm{R}_\odot$) primary radii to bracket the self-consistency range.

## 4 RESULTS

### 4.1 Orbital and dynamical solution

Table 1 summarises the observed and derived properties. The orbital-solution significance ($75.4\sigma$) and astrometric excess-noise significance ($3753\sigma$) demonstrate that the single-star model is strongly rejected, but this does not by itself validate the Keplerian fit: similar significance levels can arise from poorly resolved blends or pipeline failures. The coherent Keplerian fit and well-constrained period precision (0.27 per cent) do not exclude such explanations. *Independent RV confirmation is therefore essential.*

The semi-major axis $a = 2.64\,\mathrm{AU}$ places the companion well outside the primary's Roche lobe ($R_L \approx 0.56\,\mathrm{AU}$; filling factor $\sim$3.7 per cent), consistent with a fully detached, non-interacting system. The eccentricity $e = 0.34$ is tidally consistent: the circularisation timescale $\tau_{\mathrm{circ}} \gg \tau_{\mathrm{Hubble}}$, so the orbit retains its natal eccentricity.

The predicted RV semi-amplitude is $K_1 \approx 65\,\mathrm{km\,s^{-1}}$, readily detectable with ground-based spectrographs (Section 6.6).

### 4.2 Stellar characterisation

The extinction analysis reveals modest reddening ($A_V = 0.54\,\mathrm{mag}$). The dereddened absolute magnitude $M_G =$

1.68 mag and the photometric luminosity $L_\star = 22\,\mathrm{L}_\odot$ are consistent with the GSP-Phot evolved-star classification adopted as a working assumption, with radius $R_\star = 4.5\,\mathrm{R}_\odot$. The alternative logg-based estimates ($R = 6.5\,\mathrm{R}_\odot$, $L = 47\,\mathrm{L}_\odot$) differ by factors of $\sim 1.5\times$ (radius) and $\sim 2.1\times$ (luminosity). This is not a cosmetic discrepancy: it indicates that the stellar characterisation feeding the companion interpretation is *provisional*. The tension may reflect unreliable GSP-Phot parameters at the source's high RUWE (Section 3.4), and demonstrates that neither estimate should be treated as a secure physical measurement absent independent spectroscopy. The dereddened SED is well-fit by a single-temperature model with residuals $< 5$ per cent in all bands. No secondary SED component is required.

### 4.3 Companion-light exclusion

A main-sequence star of $M_2 = 12.3\,\mathrm{M}_\odot$ would have $L \approx 6550\,\mathrm{L}_\odot$, $R \approx 4.2\,\mathrm{R}_\odot$, and $T_{\mathrm{eff}} \approx 25\,400\,\mathrm{K}$. Using atmosphere-corrected band-by-band flux ratios (Section 3.5) with the photometric primary radius ($R = 4.5\,\mathrm{R}_\odot$), the companion-to-primary flux ratio would be $\sim 55\times$ in $G_{\mathrm{BP}}$, $\sim 29\times$ in $G$, and $\sim 18\times$ in $G_{\mathrm{RP}}$ (Table 2). Even adopting the larger logg-based radius ($R = 6.5\,\mathrm{R}_\odot$), the ratios remain $\sim 26\times$ in $G_{\mathrm{BP}}$ and $\sim 14\times$ in $G$ — overwhelmingly detectable.

The maximum main-sequence companion that could remain hidden below a 5 per cent detection threshold in all bands is $M \lesssim 1.0\,\mathrm{M}_\odot$ (photometric case) to $\sim 1.3\,\mathrm{M}_\odot$ (logg-based case). A normal main-sequence companion at the catalogue mass is therefore firmly excluded conditional on the adopted solution.

### 4.4 Mass posterior

**Caveat.** The posteriors below are *conditional working estimates*: they propagate published Gaia parameter uncertainties with an approximate external parallax-error inflation, but do *not* account for systematic errors in the `Orbital` model itself (e.g. an incorrect inclination or a spurious solution). Such errors could shift the companion mass arbitrarily and are only detectable through independent RV follow-up.

The Monte Carlo posterior (Fig. 2) yields:

**Nominal** ($\sigma_\varpi$ as published):

- Median $M_2 = 12.3\,\mathrm{M}_\odot$
- 68 per cent CI: $[8.8,\ 17.5]\,\mathrm{M}_\odot$
- 90 per cent CI: $[7.1,\ 22.4]\,\mathrm{M}_\odot$
- $P(M_2 > 5\,\mathrm{M}_\odot) = 99.7$ per cent; $P(M_2 > 10\,\mathrm{M}_\odot) = 73$ per cent

**Inflated** ($\sigma_\varpi \times 1.7$):

- Median $M_2 = 12.3\,\mathrm{M}_\odot$
- 68 per cent CI: $[7.1,\ 22.7]\,\mathrm{M}_\odot$
- 90 per cent CI: $[5.0,\ 35.9]\,\mathrm{M}_\odot$
- $P(M_2 > 5\,\mathrm{M}_\odot) = 95$ per cent; $P(M_2 > 10\,\mathrm{M}_\odot) = 64$ per cent

The wide credible intervals reflect the $\varpi^{-3}$ mass scaling (equation 3): a 10 per cent parallax error maps to a $\sim 33$ per cent mass error, and the $\times 1.7$ inflation broadens this substantially. It is important to note that the nominal posterior already has a lower 5th-percentile above $5\,\mathrm{M}_\odot$; with inflation, $\sim 5$ per cent of draws fall below the canonical BH threshold.

A parallax stress test illustrates the sensitivity directly: at $\varpi + 1\sigma$ (1.678 mas, $d = 596$ pc), the companion mass drops to $M_2 \approx 8.9\,\mathrm{M}_\odot$; at $\varpi - 1\sigma$ (1.368 mas, $d = 731$ pc), it rises to $M_2 \approx 17.5\,\mathrm{M}_\odot$.

We interpret this posterior as a conditional mass distribution under the Gaia DR3 orbital solution and adopted stellar-model assumptions, rather than as a fully independent dynamical mass measurement. These posteriors propagate only the *published parameter uncertainties* with an approximate external inflation; they do *not* account for systematic errors in the orbital model itself (e.g. an incorrectly determined inclination). Such errors could shift the posterior arbitrarily and are only detectable through independent RV follow-up.

### 4.5 Sensitivity to primary mass

Table 4 shows how $M_2$ and the BH probability vary across a grid of $M_1$ values (0.8–2.0 $\mathrm{M}_\odot$) and parallax inflation factors (1.0–2.0×). At the fiducial values ($M_1 = 1.34\,\mathrm{M}_\odot$, inflation $= 1.7$), $P(M_2 > 5\,\mathrm{M}_\odot) = 95$ per cent. For inflation factors $\geq 2.0$, the BH probability drops below 90 per cent, highlighting the importance of improved parallax precision from Gaia DR4. Fig. 3 visualises the full sensitivity landscape.

## 5 ALTERNATIVE SCENARIOS

We systematically evaluate seven non-BH explanations for the dark companion. *All verdicts in this section are conditional on the Gaia `Orbital` solution correctly describing the system and on the adopted primary-star model being approximately correct.* If the orbital solution is itself spurious (Section 6.2), these downstream exclusions lose their force. Table 3 summarises the results, using a four-tier verdict system: **Excluded** (physically impossible or overwhelmingly contradicted by data, conditional on the adopted solution), **Strongly disfavoured** (requires implausible fine-tuning, conditional on the adopted solution), **Not favoured** (unlikely but not formally excluded), and **Requires follow-up** (cannot be assessed without additional data).

**5.0.0.1 Main-sequence companion.** A $12.3\,\mathrm{M}_\odot$ MS star would have $L \sim 6550\,\mathrm{L}_\odot$ and $T_{\mathrm{eff}} \sim 25\,400\,\mathrm{K}$, contributing $\sim 29\times$ the primary flux in $G$ and $\sim 55\times$ in $G_{\mathrm{BP}}$ (atmosphere-corrected; Table 2). No secondary SED component is detected. **Excluded under adopted solution.**

**5.0.0.2 White dwarf.** $M_2 = 12.3\,\mathrm{M}_\odot$ exceeds the Chandrasekhar limit ($1.44\,\mathrm{M}_\odot$) by a factor of $\sim 9$. **Excluded under adopted solution.**

**5.0.0.3 Neutron star.** $M_2 = 12.3\,\mathrm{M}_\odot$ exceeds the most generous NS mass ceiling ($\sim 2.3\,\mathrm{M}_\odot$; Rezzolla et al. 2018) by a factor of $\sim 5$. **Excluded under adopted solution.**

**5.0.0.4 Hierarchical triple.** Two $\sim 6.2\,M_\odot$ MS stars would have combined luminosity $\sim 1160\,L_\odot$ — easily detectable in the SED. The Mardling–Aarseth stability criterion (Mardling & Aarseth 2001) would require $P_{\rm inner} \lesssim 42\,d$; no evidence for such a configuration is currently available from the existing photometric and astrometric data, though dedicated spectroscopic and time-domain observations have not yet been obtained. **Strongly disfavoured.**

**5.0.0.5 Stripped helium star / hot subdwarf.** A $12.3\,M_\odot$ stripped He star would have $T_{\rm eff} > 50\,000\,K$ and $L \sim 10^4\,L_\odot$, producing predicted NUV $\sim 9$ mag vs. observed NUV $= 17.52$ — a flux deficit of $\sim 2700\times$. No X-ray source is detected. This scenario is **not favoured**, but a cooler or lower-luminosity subdwarf cannot be fully excluded without UV spectroscopy.

**5.0.0.6 Astrometric artefact (spurious orbital solution).** Because the solution type is `Orbital` (astrometry only), there is no independent spectroscopic confirmation of the orbital parameters. El-Badry et al. (2024) have demonstrated that a fraction of high-significance `Orbital` solutions fail independent RV tests. While the $75.4\sigma$ significance, the well-constrained period, and the low GOF (18.3) do not by themselves rescue the solution from this concern: this scenario **cannot be excluded** without independent RV data. This is the single most important limitation of the present analysis.

**5.0.0.7 Chance alignment / blend-induced artefact.** At $G = 11.25$ and $b = 9.3°$, source confusion is low but not negligible. A coherent 424-d Keplerian signal at $75\sigma$ is difficult to produce from a chance blend, but scan-angle correlations with a nearby source could in principle mimic orbital motion. This scenario likewise **requires follow-up** via independent RV monitoring and high-resolution imaging.

**Summary:** Conditional on the Gaia `Orbital` solution and adopted primary-star model, three of seven alternative scenarios are inconsistent with the photometric data (MS companion, WD, NS); one (hierarchical triple) is strongly disfavoured; one (stripped He star) is not favoured but remains formally open. Crucially, two scenarios — astrometric artefact and chance alignment — *cannot be excluded* at present because the solution is purely astrometric (`Orbital`); if the solution itself is spurious, the downstream exclusions lose their force. A black-hole companion is *not excluded* by the available archival data under the adopted assumptions, but this status reflects the absence of contradicting evidence rather than positive confirmation. The system remains a *candidate* pending independent spectroscopic validation.

## 6 DISCUSSION

### 6.1 Evidence hierarchy

To clarify the epistemic status of each element of our analysis, we distinguish four tiers of evidence:

(i) **Directly measured.** Observed multi-band photometry ($G$, $G_{\rm BP}$, $G_{\rm RP}$, 2MASS, WISE, GALEX NUV), parallax ($\varpi = 1.5228 \pm 0.1549$ mas), systemic radial velocity ($v_{\rm sys} = -29.49 \pm 0.57\,{\rm km\,s^{-1}}$), and ROSAT/VLASS/RACS non-detections. These are empirical observables, subject only to standard calibration uncertainties.

(ii) **Pipeline-derived (model-dependent).** The Gaia NSS orbital parameters (period, eccentricity, angular semi-major axis, inclination) from the `Orbital` pipeline fit to along-scan astrometric residuals. The GSP-Phot stellar parameters ($T_{\rm eff}$, $\log g$), which are known to be unreliable at RUWE $= 9.31$ (Andrae et al. 2023). These quantities depend on the Gaia data reduction model and may not reflect the true system properties.

(iii) **Assumed.** The primary mass ($M_1 = 1.34 \pm 0.40\,M_\odot$) from isochrone matching of GSP-Phot parameters. The $\times 1.7$ parallax-error inflation factor, adopted from population-level validation studies (El-Badry et al. 2024) but not verified for this individual source. The uniform $[0.95, 1.05]$ multiplicative systematic on total mass. The single-temperature SED model (cross-checked against atmosphere corrections; Section 3.5).

(iv) **Conditional inference.** The companion mass posterior ($M_2 = 12.3\,M_\odot$), the dark-companion classification, and all alternative-scenario exclusions. These are valid *only if* the Gaia `Orbital` solution correctly describes the system and the adopted stellar model is approximately correct. If the orbital solution is spurious (Section 6.2) or the primary mass is substantially different from the adopted value, the entire downstream inference changes accordingly.

This hierarchy makes explicit that the paper's central result — the inferred high companion mass — sits at the fourth tier, conditional on pipeline-derived and assumed inputs that have not been independently verified for this source.

### 6.2 Reliability of the Gaia orbital solution

The reliability of the `Orbital` solution is the most consequential uncertainty in this analysis. We assess it against several available diagnostics, while emphasising that none of these constitutes a substitute for independent RV confirmation.

**6.2.0.1 Solution quality metrics.** The solution significance ($75.4\sigma$) and astrometric excess-noise significance ($3753\sigma$) indicate that the single-star model is strongly rejected. This is a necessary but not sufficient condition for a genuine orbital signal: similar significance levels arise in confirmed binaries but also in sources with poorly modelled systematics. The goodness-of-fit (GOF $= 18.3$) exceeds the ideal value of $\sim 0$, indicating that the single-Keplerian model leaves residual scatter; for context, Halbwachs et al. (2023) find GOF values of 5–30 for validated binaries at comparable RUWE, so the present value is not anomalous but also does not provide strong discriminative power between genuine and spurious solutions.

**6.2.0.2 Fraction of spurious `Orbital` solutions.** El-Badry et al. (2024) tested a sample of high-mass-ratio `Orbital` candidates against independent RV data and found that a substantial fraction ($\gtrsim$ 30–50 per cent, depending on selection criteria) showed RV semi-amplitudes inconsistent with the astrometric prediction. No systematic vetting of this specific source has been published. Applying the population-level false-positive rate as an informal prior suggests a non-trivial probability that the orbital solution does not reflect

the true system dynamics. This base-rate concern is the principal motivation for framing this source as a candidate requiring confirmation rather than a secure detection.

**6.2.0.3 Scan-angle coverage.** For `Orbital` solutions with periods near one year, the Gaia scanning law can produce near-degenerate scan-angle distributions that permit spurious orbital fits. The present system has $P = 424.4\,\mathrm{d}$ ($\sim 1.16\,\mathrm{yr}$), within the $\sim 0.8$–$1.5\,\mathrm{yr}$ window where annual scan-angle aliasing is a documented concern. Without access to the epoch-level astrometric data (not publicly available in DR3), we cannot quantify the scan-angle distribution or assess whether the orbital fit is driven by a limited range of position angles. This remains an unresolved systematic that can only be addressed with the epoch-level data expected in Gaia DR4.

**6.2.0.4 Blend probability.** At $G = 11.25$ and Galactic latitude $b = 9.3^\circ$, the local stellar density is moderate. The Gaia archive reports no resolved companion brighter than $G \sim 20$ within $2''$. However, a faint unresolved blend at sub-arcsecond separations could influence the astrometric solution. High-resolution imaging (Section 6.6) is needed to assess this possibility quantitatively.

**6.2.0.5 Assessment.** The available diagnostics are *not inconsistent with* a genuine orbital solution, but none provides positive confirmation. Given that El-Badry et al. (2024) find $\gtrsim$ 30–50 per cent of `Orbital` high-mass-ratio solutions fail independent RV tests, the prior probability that any given unvalidated `Orbital` candidate is genuine is at most $\sim$50–70 per cent before source-specific information is considered. For this source, the near-annual period (1.16 yr) adds further caution. We therefore regard the orbital solution as not excluded by available diagnostics but unverified. This assessment underlies our framing of the entire analysis as conditional and our emphasis on spectroscopic confirmation as the essential next step.

### 6.3 Why this target

The Gaia DR3 NSS catalogue contains hundreds of `Orbital` solutions with formally high companion masses, the majority of which are expected to be spurious (El-Badry et al. 2024). To justify highlighting a single source for follow-up, we quantitatively compare it against the broader `Orbital` BH-candidate population.

We extract all 47 `Orbital` sources in our working catalogue with formal companion masses $M_2 > 5\,\mathrm{M}_\odot$ and compute the predicted edge-on RV semi-amplitude

$$K_{1,\mathrm{edge}} = \left(\frac{2\pi G}{P}\right)^{1/3} \frac{M_2}{(M_1+M_2)^{2/3}} \frac{1}{\sqrt{1-e^2}} \tag{5}$$

for each. Table 5 ranks the 16 brightest ($G < 12$) among these 47 candidates by predicted $K_1$.

This source ranks **1st among 16 bright ($G < 12$) Orbital BH candidates** and 7th overall (out of 47) by predicted $K_1$. The six candidates with higher $K_1$ are all $\geq 1.2$ mag fainter, requiring significantly longer integration times for the same RV precision. Moreover, this source has the highest NSS solution significance ($75.4\sigma$) among the top ten $K_1$-ranked candidates, and its equatorial declination (Dec $= +3.5^\circ$) makes it accessible from both hemispheres.

We emphasise that *none* of these ranking criteria validate the orbital solution. They establish only that, conditional on the Gaia pipeline model, this source would be among the most efficiently confirmable BH candidates in the current catalogue. The ranking itself should be treated as a follow-up *prioritization metric*, not as evidence for a genuine BH.

### 6.4 Context among Gaia BH discoveries

Table 6 lists the orbital and stellar parameters of this system alongside the three confirmed Gaia dormant BH systems for reference. We include this comparison solely to provide context for the predicted parameter space; it should not be read as implying equivalent evidentiary standing.

There is a *fundamental evidentiary gap* between this candidate and the three confirmed systems. Gaia BH1, BH2, and BH3 each benefited from independent multi-epoch radial-velocity follow-up that validated the astrometric orbital parameters and confirmed the companion mass. In contrast, the present system has only a purely astrometric `Orbital` solution with no RV confirmation. All entries in the "This work" column of Table 6 are therefore *conditional* on the Gaia pipeline assumptions and must not be compared on equal footing with the dynamically confirmed values for BH1–3.

If the orbital solution is confirmed, the long period ($P = 424\,\mathrm{d}$) and moderate eccentricity ($e = 0.34$) would place it in a similar orbital regime to Gaia BH1 ($P = 185.6\,\mathrm{d}$, $e = 0.45$) and Gaia BH2 ($P = 1277\,\mathrm{d}$, $e = 0.52$).

### 6.5 Binary evolution implications (contingent on confirmation)

If the companion is indeed a $\sim 12\,\mathrm{M}_\odot$ BH, the progenitor was a massive star with zero-age main-sequence mass $M_{\mathrm{ZAMS}} \gtrsim 25\,\mathrm{M}_\odot$ (Fryer et al. 2012). The wide orbit ($a = 2.64\,\mathrm{AU}$) would have survived the supernova event, constraining the natal kick to moderate values. No spectroscopic abundance analysis is yet available. Independent high-resolution spectroscopy will be required to test for any abundance anomalies associated with prior mass transfer; if confirmed, the absence of significant enrichment would suggest minimal pre-collapse mass transfer, favouring a direct-collapse or failed-supernova channel (Langer et al. 2020). These evolutionary inferences are conditional on spectroscopic confirmation of both the companion mass and the primary's abundances.

### 6.6 Follow-up strategy

The predicted RV semi-amplitude $K_1 \approx 65\,\mathrm{km\,s^{-1}}$ is large, making spectroscopic confirmation feasible with modest ground-based resources. We outline the critical follow-up steps in order of priority:

(i) **Multi-epoch radial-velocity monitoring** (highest priority). At $G = 11.25$, the source is within reach of echelle spectrographs on 2–4 m class telescopes (e.g. HARPS, FEROS, CHIRON). We recommend a minimum of **8–10 RV epochs** distributed across one full orbital period ($\sim 424\,\mathrm{d}$), with phase coverage concentrated near the predicted periastron passage (where $\dot{v}_r$ is maximal). At the predicted $K_1 \approx$

$65\,\mathrm{km\,s^{-1}}$, even a precision of $\sim 0.5$–$1\,\mathrm{km\,s^{-1}}$ per epoch suffices to confirm the period, eccentricity, and semi-amplitude. Even a single well-timed epoch could reveal significant tension with the Gaia systemic velocity and motivate further monitoring, though establishing the full RV curve requires the recommended multi-epoch campaign.

The RV orbit independently yields $M_2 \sin i$ via the spectroscopic mass function; combined with the astrometric inclination, this provides a direct, pipeline-independent companion mass.

(ii) **Spectroscopic classification.** High-resolution spectroscopy would yield independent $T_{\rm eff}$, $\log g$, metallicity, detailed abundances (testing for SN-related CNO anomalies), and a spectral type — resolving the current reliance on GSP-Phot parameters.

(iii) **UV spectroscopy.** HST/COS or Swift/UVOT observations would definitively test the stripped He star scenario by searching for hot-companion UV excess.

(iv) **High-resolution imaging.** AO-assisted imaging could detect or rule out a blended optical companion responsible for a scan-angle-correlated astrometric artefact.

The target is located at RA $= 18^{\rm h}28^{\rm m}$, Dec $= +3.5°$ (Ophiuchus), accessible from both hemispheres and an evening target in the northern summer.

### 6.7 Limitations

We stress the following limitations, which collectively prevent this system from being classified as a confirmed BH:

(i) **`Orbital`-only solution.** The companion mass depends entirely on the Gaia astrometric pipeline model. No independent RV orbit exists. This is the single most important limitation and distinguishes this source from confirmed Gaia BH systems.

(ii) **Parallax-driven uncertainty.** With $\sigma_\varpi/\varpi = 10.2$ per cent and $M_{\rm tot} \propto \varpi^{-3}$, the mass uncertainty is substantial. The external parallax-error inflation ($\times 1.7$) is an approximate correction; the true systematic for this source is unknown.

(iii) **NSS covariance matrix unavailable.** Gaia DR3 does not publish the full covariance matrix of the orbital parameters. Our Monte Carlo independently samples each parameter, which may underestimate the true uncertainty if inter-parameter correlations are significant.

(iv) **GSP-Phot parameters unreliable at high RUWE.** At RUWE $= 9.31$, the primary's $T_{\rm eff}$ and $\log g$ may be biased. The $R/L$ self-consistency tension (Section 3.4) exemplifies this.

(v) **Primary mass provisional.** No independent spectroscopic mass determination exists. The adopted $M_1 = 1.34 \pm 0.40\,\mathrm{M_\odot}$ is isochrone-derived from potentially biased GSP-Phot parameters.

(vi) **Scan-angle bias possible.** For `Orbital` solutions, scan-angle correlations with nearby sources or detector-level systematics could produce spurious orbital signals. This cannot be assessed without the epoch-level astrometric data (not publicly available in DR3).

(vii) **Stripped He star not formally excluded.** The GALEX NUV non-excess and X-ray non-detections strongly disfavour this scenario but cannot definitively rule it out without UV spectroscopy.

(viii) **Approximate SED model.** Our SED check uses a single-temperature model with atmosphere corrections applied as multiplicative factors rather than a full synthetic-atmosphere fit. The MS companion exclusion is robust at the order-of-magnitude level (atmosphere corrections are 2–13 per cent; Section 3.5), but a full PHOENIX/ATLAS9 analysis would be preferable for characterising the primary and detecting faint secondary contributions.

## 7 CONCLUSIONS

We have presented an archival multi-wavelength assessment and follow-up prioritization of Gaia DR3 4277855016732107520, identifying it as a high-priority spectroscopic follow-up target from the Gaia DR3 NSS catalogue. The source carries a purely astrometric `Orbital` solution with no independent RV orbit; all inferred companion properties are conditional on the Gaia pipeline modelling assumptions, and the population-level false-positive rate for such solutions is $\gtrsim$ 30–50 per cent (El-Badry et al. 2024). Our principal findings are:

(i) **Conditional companion mass.** Under the Gaia NSS modelling assumptions and adopted stellar model, the companion mass posterior yields a median $M_2 = 12.3\,\mathrm{M_\odot}$ with 68 per cent CI $[7.1, 22.7]\,\mathrm{M_\odot}$ (with $\times 1.7$ parallax-error inflation). This posterior is a conditional working estimate: the adopted inflation factor is calibrated at the population level (not for this source), the independent parameter sampling (necessitated by the unavailability of the full NSS covariance matrix) may underestimate the true uncertainty, and the SED model is approximate. It should not be interpreted as a rigorous Bayesian posterior.

(ii) **Companion-light exclusion (conditional).** Atmosphere-corrected band-by-band flux ratios show that a main-sequence star of the required mass would exceed the primary flux by 29–55$\times$ in optical bands. This exclusion is robust at the order-of-magnitude level but rests on the adopted primary radius, which is itself uncertain (Section 4.2).

(iii) **Alternative scenarios (conditional).** Of seven non-BH scenarios, three are photometrically inconsistent with the adopted solution (MS companion, WD, NS), one is strongly disfavoured (triple), and one is not favoured (stripped He star). Two scenarios — astrometric artefact and chance alignment — *cannot be excluded* and represent the dominant source of systematic uncertainty.

(iv) **Multi-wavelength non-detections.** ROSAT non-detection implies $\log L_X \lesssim 30.7\,\mathrm{erg\,s^{-1}}$; VLASS and RACS show no radio counterpart. These are consistent with a quiescent system but do not constitute positive evidence for a BH.

(v) **Stellar characterisation provisional.** The primary's physical parameters rest on GSP-Phot values obtained at RUWE $= 9.31$, a regime where systematic biases are known. The factor-of-two luminosity tension between photometric and logg-based estimates (Section 3.4) underscores that the stellar characterisation is provisional.

(vi) **Follow-up feasibility and population ranking.** The predicted $K_1 \approx 65\,\mathrm{km\,s^{-1}}$ makes multi-epoch RV confirmation feasible with 2–4 m class facilities over $\sim$8–10 epochs spanning one orbital period (Section 6.3). Among the 16

bright ($G < 12$) `Orbital` BH candidates in our working catalogue, this source ranks first by predicted $K_1$ (Table 5).

We conclude that Gaia DR3 4277855016732107520 is a high-priority spectroscopic follow-up target: the archival data are *not inconsistent with* a massive dark companion, and the observational properties ($K_1$, brightness, accessibility) make confirmation or refutation efficient with existing ground-based facilities. However, the orbital solution is unverified, the stellar characterisation is provisional, and the statistical framework is approximate. Until independent RV monitoring confirms or refutes the orbital parameters, the companion mass and nature remain entirely provisional, and no astrophysical conclusions should be drawn from the conditional mass estimate alone.

## ACKNOWLEDGEMENTS

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## DATA AVAILABILITY

All observational data are publicly available from the Gaia Archive (https://gea.esac.esa.int/archive/), VizieR, and SIMBAD. The analysis scripts and machine-readable results are available at https://github.com/jayalabaez/dr3-4277855-black-hole-candidate. Modular analysis scripts (`scripts/02–10`) produce all figures and numerical results, including the population-level comparison (script 10; Table 5). The repository includes the exact scripts used to generate all figures, tables, and machine-readable posterior summaries from the public catalogue inputs. All numerical results are in JSON format under `results/`.

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**Table 1.** Observed and derived properties of Gaia DR3 4277855016732107520. Values marked with † depend on Gaia NSS modelling assumptions (Section 3.1).

| Property | Value | Uncertainty | Source |
|---|---|---|---|
| | *Astrometry (Gaia DR3)* | | |
| RA (J2016.0) | 274.2238° | — | Gaia DR3 |
| Dec (J2016.0) | +3.4975° | — | Gaia DR3 |
| $l, b$ | 32.12°, +9.28° | — | Gaia DR3 |
| Parallax $\varpi$ | 1.5228 mas | 0.1549 | Gaia DR3 |
| Distance | $\sim$657 pc | $\sim$67 | $1/\varpi$ |
| $G$ | 11.25 mag | 0.003 | Gaia DR3 |
| $G_{\rm BP}$ | 11.66 mag | 0.003 | Gaia DR3 |
| $G_{\rm RP}$ | 10.66 mag | 0.004 | Gaia DR3 |
| $G_{\rm BP} - G_{\rm RP}$ | 0.993 mag | — | Gaia DR3 |
| RUWE | 9.31 | — | Gaia DR3 |
| Astr. excess noise sig. | $3753\sigma$ | — | Gaia DR3 |
| | *NSS orbital solution (`Orbital`)* | | |
| Period $P$ | 424.403 d | 1.159 | Gaia NSS |
| Eccentricity $e$ | 0.3427 | 0.0194 | Gaia NSS |
| NSS significance | $75.4\sigma$ | — | Gaia NSS |
| GOF | 18.3 | — | Gaia NSS |
| $M_2$ (catalogue)† | 12.31 $M_\odot$ | 7.52 | Gaia `binary_masses` |
| | *Monte Carlo posterior (this work)* | | |
| $M_2$ (nominal)† | 12.3 $M_\odot$ | 68% CI: [8.8, 17.5]; 90% CI: [7.1, 22.4] | This work |
| $M_2$ (inflated $\times 1.7$)† | 12.3 $M_\odot$ | 68% CI: [7.1, 22.7]; 90% CI: [5.0, 35.9] | This work |
| $P(M_2 > 5\,M_\odot)$ (inflated) | 95% | — | This work |
| $P(M_2 > 10\,M_\odot)$ (inflated) | 64% | — | This work |
| | *Stellar parameters (GSP-Phot; caution: RUWE = 9.31)* | | |
| $T_{\rm eff}$ | 5922 K | $\sim$200 | Gaia DR3 |
| $\log g$ | 2.93 dex | $\sim$0.3 | Gaia DR3 |
| $v_{\rm sys}$ | $-29.49$ km s$^{-1}$ | 0.57 | Gaia DR3 |
| | *Multi-wavelength photometry* | | |
| $J$ (2MASS) | 9.84 mag | 0.02 | 2MASS |
| $H$ (2MASS) | 9.42 mag | 0.02 | 2MASS |
| $K_s$ (2MASS) | 9.33 mag | 0.02 | 2MASS |
| $W1$ (WISE) | 9.23 mag | 0.02 | AllWISE |
| $W2$ (WISE) | 9.27 mag | 0.02 | AllWISE |
| NUV (GALEX) | 17.52 mag | — | GALEX |
| | *Derived quantities* | | |
| $E(B-V)$ | 0.175 mag | 0.03 | Colour excess |
| $A_V$ | 0.543 mag | 0.09 | $R_V = 3.1$ |
| $M_G$ (dereddened) | 1.68 mag | 0.3 | Derived |
| $R_\star$ (phot.) | 4.5 $R_\odot$ | — | Photometric |
| $L_\star$ (phot.) | 22 $L_\odot$ | — | Photometric |
| $R_\star$ ($\log g$) | 6.5 $R_\odot$ | — | logg-based |
| $L_\star$ ($\log g$) | 47 $L_\odot$ | — | logg-based |
| $M_1$ (adopted) | 1.34 $M_\odot$ | 0.40 | GSP-Phot + isochrone |
| $a$ (semi-major axis) | 2.64 AU | — | Kepler III |
| $K_1$ (predicted) | $\sim$65 km s$^{-1}$ | — | Dynamics |

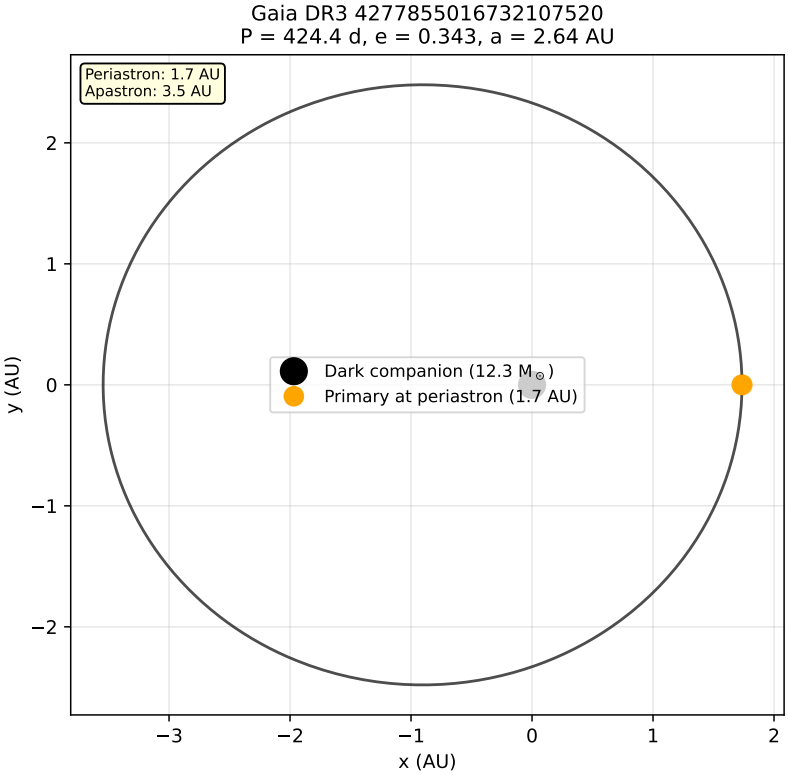


**Figure 1.** Orbital geometry of the Gaia DR3 4277855016732107520 system. The primary (GSP-Phot classification: evolved; unconfirmed) orbits the unseen companion (black circle at the focus) with period $P = 424.4\,\mathrm{d}$ and eccentricity $e = 0.34$. The periastron distance is 1.73 AU and the apastron distance is 3.55 AU.

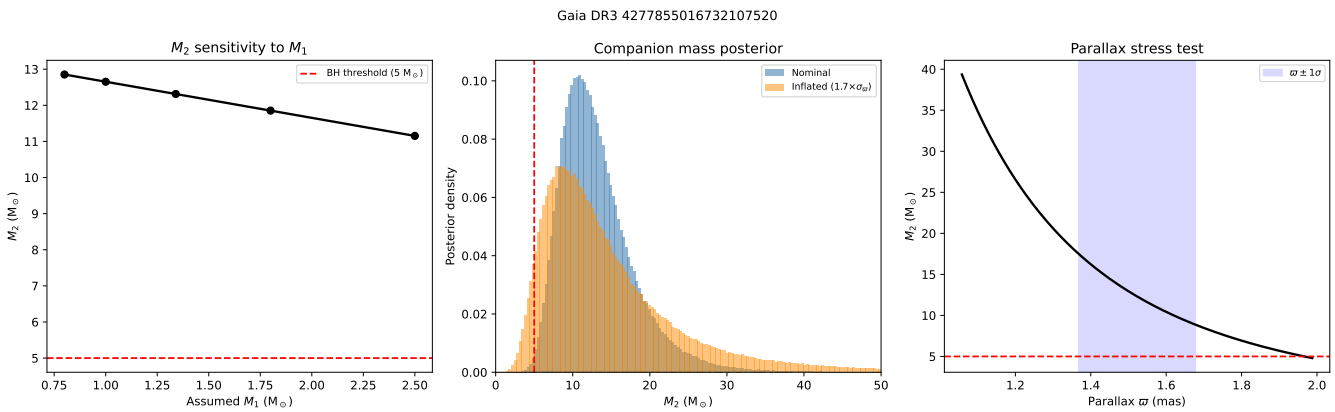


**Figure 2.** Companion mass posterior from Monte Carlo sampling. *Left:* Sensitivity to primary mass $M_1$ (all at nominal parallax uncertainty). *Centre:* Posterior with nominal (blue) and inflated ($\times 1.7$; orange) parallax uncertainty. Vertical dashed lines mark the $5\,\mathrm{M}_\odot$ and $10\,\mathrm{M}_\odot$ thresholds. The wide credible intervals reflect the $M_{\mathrm{tot}} \propto \varpi^{-3}$ scaling. *Right:* Companion mass as a function of parallax, illustrating the steep $\varpi^{-3}$ dependence. These posteriors propagate published parameter uncertainties only; they do not account for potential systematic errors in the `Orbital` model itself.

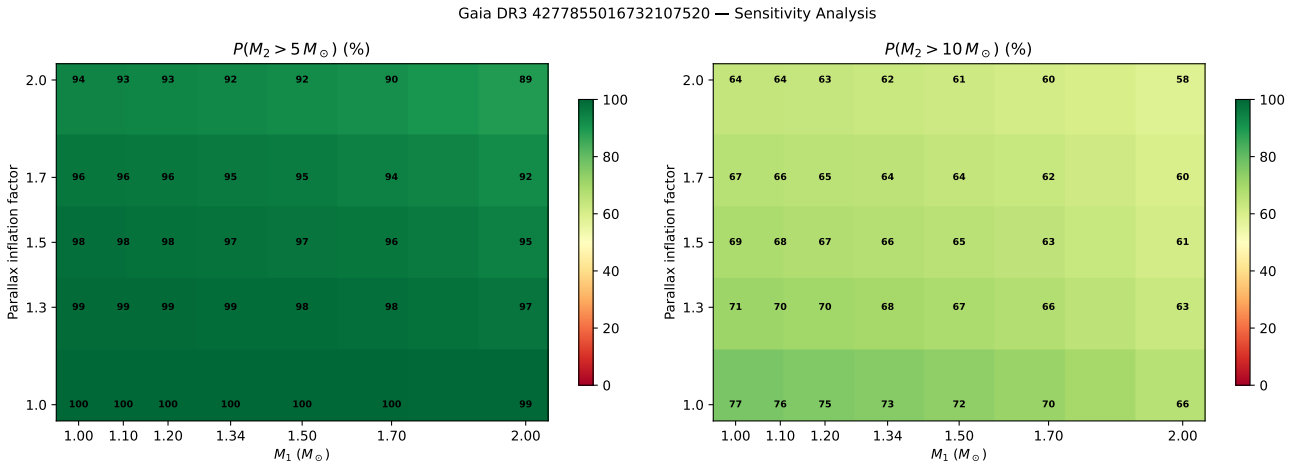


**Figure 3.** Sensitivity analysis showing $P(M_2 > 5\,\mathrm{M}_\odot)$ (*left*) and $P(M_2 > 10\,\mathrm{M}_\odot)$ (*right*) as a function of adopted primary mass $M_1$ and parallax-error inflation factor. At the fiducial values ($M_1 = 1.34\,\mathrm{M}_\odot$, inflation = 1.7), $P(M_2 > 5\,\mathrm{M}_\odot) \approx 95$ per cent.

**Table 2.** Atmosphere-corrected companion-to-primary flux ratios for a 12.3 $M_\odot$ MS companion ($T_{\rm eff}$ = 25 400 K, $R$ = 4.2 $R_\odot$) relative to the primary ($T_{\rm eff}$ = 5922 K), computed using both photometric and logg-based primary radii. Planck ratios are shown in parentheses; atmosphere corrections differ by 2–13 per cent (Section 3.5).

| Band | Phot. ($R = 4.5\,R_\odot$) | $\log g$ ($R = 6.5\,R_\odot$) |
|---|---|---|
| $G_{\rm BP}$ | 55× (49) | 26× (23) |
| $G$ | 29× (28) | 14× (13) |
| $G_{\rm RP}$ | 18× (18) | 8.4× (8.2) |
| $J$ | 8.7× (9.1) | 4.1× (4.2) |
| $H$ | 6.8× (7.1) | 3.2× (3.3) |
| $K$ | 5.8× (6.0) | 2.7× (2.8) |

**Table 3.** Alternative scenario assessment. Verdicts follow a four-tier system (Section 5). *All verdicts are conditional on the Gaia* `Orbital` *solution and adopted primary-star model.*

| Scenario | Verdict | Key constraint |
|---|---|---|
| MS companion | Excluded* | 28–49× flux excess |
| White dwarf | Excluded* | $M_2 \approx 9\,M_{\rm Ch}$ |
| Neutron star | Excluded* | $M_2 \approx 5\,M_{\rm TOV}$ |
| Triple system | Strongly disfavoured* | SED + stability |
| Stripped He | Not favoured | NUV deficit 2700× |
| Artefact | **Requires follow-up** | `Orbital` only |
| Chance align. | **Requires follow-up** | `Orbital` only |

*Conditional on the Gaia orbital solution being correct.

**Table 4.** Sensitivity of the companion mass posterior to the primary mass and parallax-error inflation factor ($\sigma_\varpi \to f \times \sigma_\varpi$). Each cell reports the median $M_2$ and $P(M_2 > 5\,\mathrm{M}_\odot)$ from $2 \times 10^5$ Monte Carlo draws.

| $M_1$ ($\mathrm{M}_\odot$) | Inflation = 1.0 | Inflation = 1.3 | Inflation = 1.7 | Inflation = 2.0 |
|---|---|---|---|---|
| 0.80 | 12.9 / 99.8% | 12.8 / 98.7% | 12.9 / 96.0% | 12.8 / 93.2% |
| 1.00 | 12.7 / 99.8% | 12.6 / 98.5% | 12.6 / 95.7% | 12.6 / 92.8% |
| 1.34 | 12.3 / 99.7% | 12.3 / 97.6% | 12.3 / 95.4% | 12.4 / 91.7% |
| 1.50 | 12.2 / 99.6% | 12.1 / 97.4% | 12.1 / 95.0% | 12.1 / 91.0% |
| 2.00 | 11.7 / 99.4% | 11.6 / 96.8% | 11.6 / 93.7% | 11.6 / 89.3% |

**Table 5.** Bright ($G < 12$) Gaia DR3 `Orbital` BH candidates ranked by predicted edge-on radial-velocity semi-amplitude $K_{1,\mathrm{edge}}$ (Equation 5). Sixteen of the 47 `Orbital` BH candidates in the working catalogue satisfy the brightness cut. $M_2$ is the pipeline-reported companion mass; $\sigma_{\mathrm{NSS}}$ the Gaia significance; "Tier" the catalogue quality flag. The target of this paper is highlighted in bold. *Rankings reflect observational accessibility and are not evidence for the reality of any individual solution.*

| Rank | Gaia DR3 source ID | $G$ (mag) | $P$ (d) | $e$ | $M_2$ ($\mathrm{M}_\odot$) | $K_{1,\mathrm{edge}}$ ($\mathrm{km\,s^{-1}}$) | $\sigma_{\mathrm{NSS}}$ | Tier | Hem. |
|---|---|---|---|---|---|---|---|---|---|
| **1** | **4277855...** | **11.25** | **424** | **0.34** | **12.3** | **65.0** | **75.4** | **Gold** | **Both** |
| 2 | 6281177... | 11.26 | 154 | 0.18 | 5.3 | 62.7 | 24.3 | Gold | Both |
| 3 | 4507516... | 8.89 | 813 | 0.51 | 18.2 | 48.1 | 20.3 | Gold | Both |
| 4 | 6042980... | 11.96 | 678 | 0.22 | 9.1 | 47.8 | 76.3 | Gold | Both |
| 5 | 6033351... | 11.75 | 857 | 0.41 | 6.9 | 41.7 | 55.5 | Gold | Both |
| 6 | 3153895... | 9.34 | 904 | 0.00 | 16.6 | 41.7 | 18.1 | Silver | Both |
| 7 | 245948... | 11.80 | 549 | 0.07 | 5.0 | 35.1 | 48.4 | Gold | N |
| 8 | 6656986... | 6.47 | 444 | 0.58 | 5.5 | 31.6 | 28.0 | Gold | S |
| 9 | 2946055... | 5.27 | 755 | 0.32 | 11.0 | 30.6 | 14.6 | Silver | Both |
| 10 | 6886105... | 9.46 | 768 | 0.48 | 6.9 | 30.0 | 34.9 | Gold | Both |
| 11 | 4113903... | 6.81 | 1233 | 0.19 | 7.3 | 25.9 | 6.1 | Bronze | Both |
| 12 | 3067772... | 11.64 | 984 | 0.17 | 5.7 | 25.3 | 20.8 | Silver | Both |
| 13 | 3972084... | 8.80 | 876 | 0.15 | 6.1 | 25.0 | 21.0 | Gold | Both |
| 14 | 979031... | 9.94 | 877 | 0.22 | 5.1 | 22.5 | 18.4 | Bronze | N |
| 15 | 470361... | 7.18 | 999 | 0.52 | 6.5 | 21.7 | 11.2 | Silver | N |
| 16 | 5103956... | 6.88 | 763 | 0.27 | 5.0 | 20.8 | 22.5 | Gold | Both |

**Table 6.** Comparison with confirmed Gaia dormant BH systems (included for contextual reference only). Gaia BH1–3 each have independent multi-epoch RV orbits confirming the companion mass; this source has only a purely astrometric `Orbital` solution with no RV confirmation. The two classes of evidence are not comparable, and the "This work" column reflects conditional inferences under the Gaia pipeline assumptions.

| Property | Gaia BH1 | Gaia BH2 | Gaia BH3 | This work |
|---|---|---|---|---|
| $P$ (d) | 185.6 | 1277 | ∼4240 | 424.4 |
| $e$ | 0.45 | 0.52 | 0.73 | 0.343 |
| Primary type | G-dwarf | RGB | Giant | GSP-Phot evolved[a] |
| $M_1$ ($\mathrm{M}_\odot$) | 0.93 | 1.07 | 0.76 | 1.34 |
| $M_2$ ($\mathrm{M}_\odot$) | 9.62 | 8.94 | 32.70 | ∼12.3[b] |
| Solution type | Astro+RV | Astro+RV | Astro+RV | `Orbital` only |
| RV confirmation | Yes | Yes | Yes | **No** |
| $d$ (pc) | 480 | 1160 | 590 | ∼657 |
| $G$ (mag) | 13.8 | 12.3 | 11.6 | 11.25 |
| Status | Confirmed | Confirmed | Confirmed | **Candidate** |

[a] From GSP-Phot only (RUWE = 9.31); no independent spectral classification.
[b] Pipeline-derived mass under Gaia NSS assumptions; 68% CI [7.1, 22.7]; 90% CI [5.0, 35.9] with ×1.7 inflation.